Project and Professionalism (6CS007)

Professionalism Report - AI Powered Resume Generator

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1. Introduction

In Software Engineering, being professional does not just mean producing software that operates correctly it also includes being ethical and socially conscious as well as complying with applicable legislation and looking after the interests of users during all stages of a system's lifecycle (Sommerville, 2016). Software Engineers should also consider the impact their systems use will have on others (individuals, businesses, society, etc.), particularly when their applications will impact an individual's ability to obtain a job, gain access to private information and the opportunities available for personal growth (Pressman & Maxim, 2020). Thus, to be considered a professional, Software Engineers must possess technical skills however, they must also demonstrate accountability, honesty and adherence to standards and best practices.

This paper examines the professionalism aspects of an AI Powered Resume Generator which is a web based application that helps people seeking employment create resumes, optimize resumes using AI, match their resume content to job postings and prepare for Mock Interviews. The AI Powered Resume Generator uses advanced technology such as artificial intelligence (AI) with the latest web technologies to enhance outplacement results as documented in the related artefact. The AI Powered Resume Generator is designed to process personal and professional information and provide AI based feedback to users which could ultimately affect an individual's career trajectory, additional ethical, social, legal and security issues need to be taken into account when developing the platform (Sommerville, 2016).

The purpose of this report is to evaluate the project using a critical perspective based on professional software engineering practices rather than just technical aspects of implementation. To do this, the report will look at the project in terms of its social effects, ethical responsibilities, legal responsibilities and security protocols as supported by existing academic research and industry recommendations. Currently, the project is an academic prototype however, this report will evaluate the project's effectiveness against professional standards and highlight the factors that need to be taken into account when deploying the project correctly in the real world (Pressman & Maxim, 2020).

2. Project Overview

The AI Powered Resume Generation Application is a modular web based application created to support users through their entire employment preparation process. Users will be able to register & securely authenticate their account, create a resume using structured templates, gain AI driven feedback from their resume based on the job description they want to apply to, identify gaps in their skill set and practice answering interview questions by accessing practice interviews (mock interviews) generated by an AI program. These capabilities will be provided in multiple distinct modules including User Management, Resume Builder, AI Optimization and Job Matching and Mock Interview Preparation Modules.

This project uses an Agile Project Development Approach supported by Professional Artefacts such as product backlogs, sprint backlogs, Functional Decomposition Diagrams (FDD), Unified Modeling Language (UML) Diagrams, Wireframe and Structured Testing Strategies. The use of these artefacts improves communication between Stakeholders, decreases ambiguity and allows for iterative improvement of the product, which are core principles of any professional Software Engineering Practice (Schwaber & Sutherland, 2020) (Sommerville, 2016).

The use of AI to analyze and match resumes also creates new professional responsibilities as a result of reliance on AI. The way users view their skill sets and employability will be impacted by the influence of AI on the process by which they prepare applications and are matched with open positions. Sommerville states that any system that provides support for or influences human decision making should consider the ethical risks associated with the use of an AI system, as well as the level of transparency and ability to mitigate potential unintended consequences. Thus, professional responsibility for this project not only lies with technical correctness but also with the level of responsible design and communication to end users regarding AI related features.

3. Social Impact

3.1. Positive Social Impact

A primary benefit of the AI Powered Resume Generator is that it allows all students, graduates and job seekers to gain access to resume writing services, career coaching and interview preparation at no cost. Traditionally, these services are expensive and only available to a few wealthy individuals. Mainly, the Resume Generator eliminates the barriers that prevent individuals from preparing their resumes and matching themselves to job opportunities, thereby creating a larger diversity of individuals who are able to successfully pursue their professional goals. The Resume Builder feature has been created to provide a user friendly means for users to create a professional grade resume using a variety of professionally formatted templates. Users can create these resumes and receive a real time preview while they build their resumes. This approach makes it easier for users who may not have a lot of design or technical skills to use, as well as increases the usability of this module for users who have varying degrees of digital literacy. Research shows that user centered design is associated with better accessibility and lower cognitive load, which leads to higher engagement and satisfaction with the user experience (Garrett, 2011).

The AI Optimization and Job Matching module contributes to the positive social impact that the AI Powered Resume Generator will have by helping users determine the types of skills and qualifications that employers in specific industries expect employees to possess. The module also identifies skills that users may be lacking and keywords related to those skills that they will need to include in their resumes to be competitive for positions in those industries (Sommerville, 2016). This effectively supports society's pursuit of developing a workforce that is ready for work and will further increase their employability as a result.

Mock Interview Preparation allows users to improve their self-esteem when preparing for job interviews by practicing interviews using simulations. Using simulated interviews can lead to higher confidence levels for people who do not have a job network or have never interviewed before. Building employability and confidence through the Mock Interview Preparation is an example of a socially responsible approach to designing a system, because it demonstrates that the opportunity to obtain employment is directly related to one's economic stability and quality of life (Pressman & Maxim, 2020).

3.2. Potential Negative Social Impact

Although there are many advantages to implementing the system, it could create potential adverse social consequences if implemented improperly. For example, it may lead to an increased dependence on AI generated feedback, which could result in users viewing the AI recommendations as absolutes or infallible, thus reducing the likelihood of independent judgment via another source. In addition to that, Professional Software Systems should enhance the ability to evaluate using logical and analytical approaches rather than replacing them. This is especially important in areas where there may be subjective opinions or judgement involved (e.g., hiring decisions) (Pressman & Maxim, 2020).

Another issue that may arise is Digital Exclusion. Individuals who do not have reliable internet access or access to appropriate devices and or adequate digital skill levels will not be able to fully utilize the platform if they do not possess these resources. This lack of access to technology could exacerbate the existing inequalities associated with access to technology, which is a challenge recognized in the field of Digital Service Design (Sommerville, 2016).

The implementation of Automated Resume Scoring and Job Matching has the potential to inadvertently propagate existing biases embedded within training data (e.g., Job Descriptions) or to amplify the existing biases introduced into formulation processes. Unchecked, these biases may adversely impact specific demographic groups, leading to a reduction of equitable access to opportunities for employment or an increased risk of being hired based on an AI generated assessment, rather than direct experience and or education. Through the above, it becomes apparent that a professional understanding of the overall social impact of implementing systems that are driven by artificial intelligence (AI) is required (Sommerville, 2016).

4. Ethical Considerations

4.1. User Autonomy and Fairness

The principles of ethical software engineering promote the maintenance of user autonomy and fairness in the development and use of software systems. In support of user autonomy, the AI Powered Resume Generator enables users to choose whether and how to use any recommendations generated by the AI system. There is no automated submission of resumes or decisions regarding employment made on behalf of users by the system therefore, users maintain control over their own employment opportunities. Preserving user agency is an important professional ethical principle addressed in the design of a professional system (Sommerville, 2016, p. 11).

The principle of fairness is expressed in that all users will be provided with the same level of feedback regarding their resumes and that the AI system will not generate reports that are discriminatory against specific groups of users based upon their race, ethnicity, gender identity or disability. Professional engineers have an ethical responsibility to evaluate the AI generated data continuously to ensure that they do not provide a disproportionate advantage or disadvantage for a particular group of users. In addition to the functionality of the AI system, professional engineers should consider the fairness of each outcome produced by the AI system (Pressman & Maxim, 2020, p. 94).

4.2. Bias, Transparency and AI Ethics

The ethical implications of AI systems include the following issues: bias, accountability, transparency. Resume analysis algorithms (and job matching algorithms) have the potential to reflect the bias that exists within historical employment practices or in the training data. As part of implementing ethical designs within AI, designers must acknowledge the ethical impacts and provide strategies/processes that mitigate the risks associated with using AI generated content. Strategies/processes such as providing users with a transparent understanding of what they are using and providing a level of human oversight over the decision making process (Sommerville, 2016).

One of the most important aspects of transparency is building trust from users of the AI system. Users need to understand that AI generated resumes have probabilistic outcomes depending on several contextual factors therefore, users should not rely solely on AI generated resume scores/recommendations as an objective measure of employability. By failing to communicate the limitations of the system's capabilities, users may misinterpret or misuse the information provided by the AI system, which constitutes an ethical issue (Myers, 2011).

4.3. Privacy, Consent and Data Responsibility

Personal information is processed by a system to include resumes, employment history and user profile information. Minimizing data collected, securing all stored information and making sure that users give consent prior to using their information are all ethical responsibilities. The artefact recognizes these ethical responsibilities through the use of secure authentication, role based access control and restricting access to certain administrative privileges.

The ethical handling of personal career-related data is paramount, as the unintentional abuse of such data or exposure of it could damage a user's career. Therefore, all software engineers who develop or work with software must be fully informed about proper data practices therefore, all software must follow the rules of ethical data management (Pressman & Maxim, 2020).

4.4. Ethical Limitations and Risk Mitigation

The ethical restrictions around the system, which can be considered an academic prototype, have not been properly accounted for because there is no formal bias audit or explain ability for the system and its AI outputs. It demonstrates maturity and ethics on the part of the authors to acknowledge this ethical shortcoming. Future development of the system should involve human in the loop validation, bias testing and adding clearer disclaimers for users to lower the ethical risk associated with using an unlimited system and provide more trust in it (Sommerville, 2016).

5. Legal Considerations

5.1. Data Protection and GDPR Alignment

Although the system was developed for educational purposes, the principles of the European Union's GDPR have been integrated into this system specifically, the GDPR defines the following principle based guidelines: ‘data minimization’, ‘limitation of purpose’ and ‘restricted access to personal data’. A resume is classified as personal data and many times the resume will contain sensitive types of data therefore, it must be handled properly to prevent any unintended or unauthorized release of that data (Sommerville, 2016).

There is a level of professional foresight demonstrated when designing a system with regulatory compliance in mind, even if not specifically mandated by law. Account for legal concerns while designing the system, thus lowering the chance of encountering any future technical and legal risks (Pressman & Maxim, 2020).

5.2. Intellectual Property and Licensing

This project shows knowledge of legal frameworks by utilizing open source software frameworks and technologies that have the right licenses, including React.js and Node.js. By adhering to software licenses and intellectual property rights, you can avoid lawsuits and ethical violations, which is a key part of being a professional software engineer (Pressman & Maxim, 2020).

5.3. User Rights and Accountability

Professional systems need to provide users with access to their own data, the ability to correct incorrect information and the ability to delete their data. Although the prototype does not include these functionalities at this time, the architecture of the Professional System does support adding these capabilities in the future. Acknowledging users' rights is a legal and ethical responsibility to design a system (Sommerville, 2016).

6. Security Considerations

6.1. Authentication and Access Control

It is very important to ensure the security of systems that manage sensitive personal information, so implementing secure methods for authenticating users is vital. To protect your users against unauthorized access and to comply with the latest industry standards (Pressman & Maxim, 2020). The AI Powered Resume Generator has implemented secure authentication techniques such as: JWTs (JWTs provide security token based sessions), Password Hashing (hashed passwords are stored rather than storing user's password in plain text) and Role Based Access Control.

6.2. Protection of Sensitive User Data

Controlled API's (Backend) and limited access to administrator accounts help ensure that user profiles and resumes are secure. Access Control reduces data breach risks or accidental exposure. A significant amount of security incidents occurs due to a lack of access restrictions rather than sophisticated attacks thus, the importance of having basic security controls in place (Hamilton, 2021).

6.3. System Reliability and Integrity

A critical function of system reliability is to create user confidence in the software product created (Sommerville, 2016). Although the prototype demonstrated proof of operating consistently, users will still need to have continual supportive activities such as monitoring, backup procedures and fault tolerance capabilities to successfully utilize the prototype within a production environment. Furthermore, with respect to professional applications of software, the ability of the system to perform reliably, is one of the most significant quality characteristics of an effective software system.

7. Professional Software Engineering Practice

Demonstrating a commitment to Professional Software Engineering Standards is supported by using many different software development artefacts such as UML diagrams, Functional Decomposition Diagrams (FDD), Wireframes and Test Cases. These artefacts provide a means for all stakeholders to have a clear understanding of the project from beginning to end, as well as providing for structured development and assurance of quality throughout the lifecycle of the software (Sommerville, 2016). Embracing the Agile Way of Working also reflects industry best practices by utilizing iterative and Incremental Development, Stakeholder Collaboration and Continuous Improvement (Schwaber & Sutherland, 2020).

8. Conclusion

The AI Powered Resume Generator was developed with consideration for professional software engineering responsibilities however, it also supports social needs through increasing accessibility to tools used in career development as well as recognizing the ethical risks of using artificial intelligence (AI) to drive decision making. In conjunction with professional software development responsibilities these additional legal, security and development support structures augment the project with a sound professional base.

While the current prototype version of the system has been created, it serves as a foundation for future product range enhancements. Potential enhancements that could lead to safe and responsible use of AI Powered Resume Generator in a production environment include bias auditing, enhanced privacy controls and a formal compliance process with legal requirements. All aspects of the project are consistent with the principles of professionalism in Software Engineering espoused by (Sommerville, 2016) and (Pressman & Maxim, 2020).

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